

Artificial Intelligence Research Notes

This document is intended as a realistic sample PDF for testing PDF parsing, chunking, embeddings, semantic search, and retrieval in an AI assistant project.

Introduction

Artificial Intelligence (AI) is the field of building systems capable of performing tasks that normally require human intelligence. Examples include language understanding, computer vision, recommendation systems, and robotics.

Machine Learning

Machine learning allows computers to learn patterns from data rather than relying on explicit rules. Supervised learning uses labeled data, unsupervised learning discovers hidden structure, and reinforcement learning learns through rewards.

Large Language Models

Large Language Models (LLMs) are trained on massive text corpora using transformer architectures. They generate text, answer questions, summarize documents, and assist with programming.

Retrieval-Augmented Generation

RAG combines semantic search with language models. Documents are split into chunks, converted into vector embeddings, stored in a vector database, and retrieved based on similarity before generating an answer.

Example Data

Important technologies include Node.js, Express, PostgreSQL, MongoDB, Redis, Docker, Kubernetes, FastAPI, LangChain, OpenAI APIs, and vector databases like Qdrant or Pinecone.

Conclusion

A production AI assistant typically includes authentication, document upload, OCR, chunking, embedding generation, vector storage, retrieval, prompt construction, and answer generation.

Sample Table (plain text)

Model	Parameters	Primary Use
GPT-4	Large	General reasoning
Llama 3	Various	Open-source applications
Mistral	Medium	Efficient inference

Keywords: embeddings, cosine similarity, chunk overlap, OCR, metadata, semantic search, vector database, BM25, hybrid search, reranking.